B. pSort

time limit per test: 2 seconds
memory limit per test: 256 megabytes

One day n cells of some array decided to play the following game. Initially each cell contains a number which is equal to it's ordinal number (starting from 1). Also each cell determined it's favourite number. On it's move i-th cell can exchange it's value with the value of some other j-th cell, if $|i - j| = d_i$, where d_i is a favourite number of i-th cell. Cells make moves in any order, the number of moves is unlimited.

The favourite number of each cell will be given to you. You will also be given a permutation of numbers from 1 to n. You are to determine whether the game could move to this state.

Input

The first line contains positive integer n ($1 \le n \le 100$) — the number of cells in the array. The second line contains n distinct integers from 1 to n — permutation. The last line contains n integers from 1 to n — favourite numbers of the cells.

Output

If the given state is reachable in the described game, output `YES`, otherwise `NO`.

Examples

input Copy

```
5
5 4 3 2 1
1 1 1 1 1
```

output Copy

```
YES
```

input Copy

```
7
4 3 5 1 2 7 6
4 6 6 1 6 6 1
```

output Copy

```
NO
```

input Copy

```
7
4 2 5 1 3 7 6
4 6 6 1 6 6 1
```

output Copy

```
YES
```

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 28 (Codeforces format)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msys2

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
372684287	Apr/27/2026 11:30	Accepted
372683875	Apr/27/2026 11:24	Accepted

→ Problem tags

dfs and similar | dsu | graphs | *1600

No tag edit access

→ Contest materials

- Announcement
- Tutorial #1 (en)
- Tutorial #2
- Tutorial #3 (ru)
- Tutorial #4
- Tutorial #5

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